

EXPERIENCE

Cloudflare

Senior Application Security Engineer · 2023-Present

- Led security review of microservices platform processing 25M+ requests/sec; identified 18 critical vulnerabilities reducing CVSS exposure by 94% and preventing estimated \$12M+ in breach costs
- Architected SAST/DAST pipeline reducing mean time to vulnerability detection from 23 days to 4.2 days across 180+ engineering teams; achieved 99.2% false positive suppression rate
- Designed custom WAF rules for GraphQL APIs blocking 2.3M+ attack payloads monthly (99.7% precision) while maintaining 99.95% legitimate traffic throughput

Stripe

Application Security Engineer · 2021-2023

- Established threat modeling framework adopted across 12 product teams; completed 47 threat models discovering 156 design-level risks with 89% remediation rate within SLA
- Reduced payment API latency from 18ms to 12.4ms (31% improvement) while adding comprehensive input validation; handled \$890B+ in annual transaction volume with zero authentication bypasses
- Built automated secrets detection system scanning 2.1M commits/month across monorepo; prevented 23 incidents preventing exposure of AWS/payment processor credentials

Wise

Security Engineer (Application Security Focus) · 2019-2021

- Led incident response for 8 critical vulnerabilities in payment reconciliation services affecting 4.2M+ user accounts; achieved 100% remediation in 6.5 hours average MTTR
- Implemented OAuth 2.0/OIDC authentication overhaul serving 180+ API clients; reduced authentication-related incidents by 87% while decreasing token validation latency by 15.3ms
- Conducted 52 secure code reviews across fintech Python/Golang codebases; trained 34 engineers achieving 78% reduction in OWASP Top 10 findings in subsequent sprints

Palantir Technologies

Security Intelligence Analyst (Contract) · 2018-2019

- Analyzed 340+ data pipeline configurations identifying 67 privilege escalation pathways; remediation roadmap prevented estimated 12 high-impact

PROJECTS

VulnRadar

Python, FastAPI, PostgreSQL, Kubernetes, YARA Rules · 2022-Present

- Open-source runtime vulnerability scanner for containerized applications; 2,840+ GitHub stars, 280K+ monthly downloads; detects 340+ CVE patterns with 92.3% true positive rate
- Implemented adaptive ML-based anomaly detection reducing scan time by 68% (from 240s to 77s per image) while maintaining 99.1% vulnerability coverage accuracy

CryptoShield

Rust, C, LLVM, OpenSSL, TLS 1.3 · 2021-2023

- Cryptographic vulnerability detection library for C/C++/Rust codebases; 1,240+ stars, 95K+ weekly downloads; identified 156+ real-world cryptographic misuse patterns in production systems
- Achieved sub-2ms analysis latency per 10K lines of code with zero false negatives on 45 OWASP crypto test cases; integrated into CI/CD pipelines for 40+ enterprises

AuthPulse

TypeScript, Node.js, Next.js, OAuth 2.0, SAML 2.0 · 2020-2022

- Reference implementation and testing harness for OAuth 2.0/OIDC compliance; 1,680+ GitHub stars, 156K+ total clones; certified against 34 official OpenID Foundation test cases
- Provides 18+ pre-configured vulnerability scenarios with 99.7% accuracy in simulating real-world authentication attacks; used in security training by 12 Fortune 500 companies

EXTRACURRICULAR

Black Hat USA & OWASP Global AppSec

Conference Speaker & Training Instructor · 2021-Present

- Delivered 8 talks on microservices security and API threat modeling to 2,400+ security practitioners; 2023 talk 'Zero-Trust Architecture for Polyglot APIs' achieved 4.8/5.0 attendee rating
- Designed 6-hour intensive workshop on runtime security and behavior-based anomaly detection; trained 340+ engineers with 89% reporting significant improvement in threat detection capabilities

OWASP Project Leadership & OSS Mentorship

OWASP Top 10 Maintainer & Senior Mentor (Linux Foundation) · 2020-Present

- Co-maintained OWASP Top 10 2021 release guiding 45+ security researchers;

EDUCATION

Imperial College London

MSc Cyber Security · 2019-2021

- Achieved distinction with 78% average; led secure coding workshop for 45+ students covering OWASP Top 10 and threat modeling
- Thesis on 'Runtime API Monitoring for Zero-Day Detection in Microservices' published as conference paper with 340+ citations

University of Waterloo

BASc Computer Science (Honours) · 2015-2019

- Graduated with 3.87/4.0 GPA; President of Cybersecurity Club growing membership from 12 to 120+ members over 2 years
- Co-authored 2 peer-reviewed security papers on cryptographic side-channel attacks and supply chain vulnerability analysis

SKILLS

Languages

Python Rust Go TypeScript
Java C SQL Bash

Technologies

OAuth 2.0 OIDC SAML 2.0
TLS 1.3 AES-256 RSA WebAuthn
FIDO2 Kubernetes Docker
Apache Kafka gRPC GraphQL
REST APIs JWT

Tools

Burp Suite Professional OWASP ZAP
SonarQube Snyk Checkmarx
GitLab SAST AWS Security Hub
Datadog ELK Stack Splunk
Threat Modeling (Microsoft Threat Modeling Tool)
Wireshark Metasploit

incidents based on threat intelligence correlation

- Developed Python framework for automated SQL injection pattern detection across distributed systems achieving 94.2% recall with 98.1% precision on 180K+ query samples
- Conducted red team assessment of access control matrices across 18 internal platforms; discovered 23 logic flaws including IDOR and broken function-level access control allowing lateral movement

document received 2.3M+ downloads and integrated into 180+ university cybersecurity curricula globally

- Mentored 12 junior security engineers through formal program; 11 mentees promoted to senior/lead roles within 18-24 months with average 34% salary increase